

Example 3: A string of 10 light bulbs is connected in series, which means that the entire string will not light up if any one of the light bulbs fails. Assume that the lifetimes of the bulbs, $\tau_1, \tau_2, \dots, \tau_{10}$, are independent random variables that are exponentially distributed with mean 1/2. Find the distribution of the life length of this string of light bulbs.

Solution: Note that the pdf of τ_i is $f(t) = 2e^{-2t}$, $0 < t < \infty$, and the cumulative distribution of τ_i is $F_{\tau_i}(t) = 1 - e^{-2t}$, $t > 0$.

Let T represent the lifetime of this string of light bulbs. Then, $T = \min(\tau_1, \tau_2, \dots, \tau_n)$. Thus,

$$F_T(t) = 1 - [1 - F_{\tau_i}(t)]^{10}$$

Hence, the density of T is obtained by differentiating $F_T(t)$ with respect to t , that is,

$$\begin{aligned} f_T(t) &= 10f_{\tau_i}(t)[1 - F_{\tau_i}(t)]^9 \\ &= \begin{cases} 2(10)e^{-2t}(e^{-2t})^9, & 0 < t < \infty \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} 20e^{-20t}, & 0 < t < \infty \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

The joint pdf of the order statistics is given by the following result.